

# Optimisation of Multiple Control Strategies on a Dengue Fever Model using JuMP.

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## Introduction

This example explores the optimal control on a dengue fever model using multiple combinations of NPIs strategies. The model and strategies tested match those presented by [Asamoah et al. \(2021\)](#). The model takes into account 2 populations: human and female mosquito or vector population. The human population is divided into 5 compartments: susceptible  $S_h$ , infected (symptomatic)  $I_h$ , carrier (asymptomatic)  $I_hA$ , partially immune  $P$ , and recovered  $R_h$ . While the mosquito population is formed by 2 compartments: susceptible  $S_v$  and infected  $I_v$ .

The model is described by the following differential equations:

$$\begin{aligned}N_h(t) &= S_h(t) + I_h(t) + I_hA(t) + P(t) + R_h(t), \\N_v(t) &= S_v(t) + I_v(t), \\ \lambda_h(t) &= \frac{(1 - u_1(t))b\beta_1}{N_h(t)}I_v(t), \\ \lambda_{h1}(t) &= \frac{(1 - u_1(t))b\beta_2}{N_h(t)}I_v(t), \\ \lambda_v(t) &= \frac{b\beta_3}{N_h(t)}(I_h(t) + I_hA(t)).\end{aligned}$$

$$\begin{aligned}
\frac{dS_h}{dt} &= \mu_h N_h - \lambda_h S_h - S_h u_2 - \mu_h S_h, \\
\frac{dI_h}{dt} &= \psi \lambda_h S_h + \omega \lambda_h 1P - (\mu_h + u_3 + \gamma_h) I_h, \\
\frac{dI_h A}{dt} &= (1 - \psi) \lambda_h S_h + (1 - \omega) \lambda_h 1P - (\mu_h + \gamma_h) I_h A, \\
\frac{dP}{dt} &= u_2 S_h + \rho u_3 I_h + \phi \gamma_h (I_h + I_h A) - \lambda_h 1P - \mu_h P, \\
\frac{dR_h}{dt} &= (1 - \rho) u_3 I_h + (1 - \phi) \gamma_h (I_h + I_h A) - \mu_h R_h, \\
\frac{dS_v}{dt} &= \mu_v N_v (1 - u_4) - \lambda_v S_v (1 - u_1) - \mu_v S_v - r_0 u_4 S_v, \\
\frac{dI_v}{dt} &= \lambda_v S_v (1 - u_1) - \mu_v I_v - r_0 u_4 I_v.
\end{aligned}$$

## Libraries

```

using JuMP
using Ipopt
using OrdinaryDiffEq
using Plots
using Measures
using MAT

```

## Functions

Model

```

function dengue_ode!(du,u,p,t)
    (S_h, I_h, I_hA, P, R_h, S_v, I_v) = u
    (β1, β2, β3, ρ, ψ, γ_h, ω, μ_h, μ_v, φ, r_0, u1, u2, u3, u4) = p
    N_h = S_h + I_h + I_hA + P + R_h
    N_v = S_v + I_v
    λ_h = ((1 - u1) * b * β1 / N_h) * I_v
    λ_h1 = ((1 - u1) * b * β2 / N_h) * I_v
    λ_v = (b * β3 / N_h) * (I_h + I_hA)

```

```

@inbounds begin
    du[1] =  $\mu_h * N_h - \lambda_h * S_h - S_h * u2 - \mu_h * S_h$ 
    du[2] =  $\psi * \lambda_h * S_h + \omega * \lambda_{h1} * P - (\mu_h + u3 + \gamma_h) * I_h$ 
    du[3] =  $(1 - \psi) * \lambda_h * S_h + (1 - \omega) * \lambda_{h1} * P - (\mu_h + \gamma_h) * I_{hA}$ 
    du[4] =  $u2 * S_h + \rho * u3 * I_h + \phi * \gamma_h * (I_h + I_{hA}) - \lambda_{h1} * P - \mu_h * I_h$ 
    du[5] =  $(1 - \rho) * u3 * I_h + (1 - \phi) * \gamma_h * (I_h + I_{hA}) - \mu_h * R_h$ 
    du[6] =  $\mu_v * N_v * (1 - u4) - \lambda_v * S_v * (1 - u1) - \mu_v * S_v - r_0 * I_v$ 
    du[7] =  $\lambda_v * S_v * (1 - u1) - \mu_v * I_v - r_0 * u4 * I_v$ 
end
nothing
end;

```

To create plots

```

function create_combined_plot(ts, u_opts, S_h_opts, I_h_opts, I_hA_opts, P_opts,
    color_palette = [:red, :blue, :green, :magenta, :purple, :cyan, :orange, :yellow],
    linestyle_palette = [:solid, :dash, :dot, :dashdot, :dashdotdot, :dash, :dot])

    combined_plot = plot(layout = (3, 2), dpi=300, size=(1200,1500),
        left_margin=10mm, right_margin=10mm, top_margin=10mm)

    # First subplot:
    for (i, I_h_opt) in enumerate(I_h_opts)
        plot!(combined_plot[1, 1], ts, I_h_opt,
            label="Scenario $(scenario_label[i])",
            linewidth=3, color=color_palette[i], thickness_scaling=1,
            xlim=(0, ts[end]),
            ylim=(0, 600),
            xtickfontsize=12, ytickfontsize=12,
            xguidefontsize=14, yguidefontsize=14,
            legendfontsize=12, legend=:topright)
    end
    xlabel!(combined_plot[1, 1], "Time (days)")
    ylabel!(combined_plot[1, 1], "Symptomatic humans, I_h")
    title!(combined_plot[1, 1], "Symptomatic humans, I_h")

    # Second subplot:
    for (i, I_hA_opt) in enumerate(I_hA_opts)
        plot!(combined_plot[1, 2], ts, I_hA_opt,
            label="Scenario $(scenario_label[i])",
            linewidth=3, color=color_palette[i], thickness_scaling=1,

```

```

        xlim=(0, ts[end]+1),
        ylim=(0, 800),
        xtickfontsize=12, ytickfontsize=12,
        xguidefontsize=14, yguidefontsize=14,
        legendfontsize=12, legend=:topright)
    end
    xlabel!(combined_plot[1, 2], "Time (days)")
    ylabel!(combined_plot[1, 2], "Asymptomatic humans, I_hA")
    title!(combined_plot[1, 2], "Asymptomatic humans, I_hA")

    # Third subplot:
    for (i, S_v_opt) in enumerate(S_v_opts)
        plot!(combined_plot[2, 1], ts, S_v_opt,
            label="Scenario $(scenario_label[i])",
            linewidth=3, color=color_palette[i], thickness_scaling=1,
            xlim=(0, ts[end]),
            ylim=(0, 6500),
            xtickfontsize=12, ytickfontsize=12,
            xguidefontsize=14, yguidefontsize=14,
            legendfontsize=12,
            legend=:right)
    end
    xlabel!(combined_plot[2, 1], "Time (days)")
    ylabel!(combined_plot[2, 1], "Susceptible mosquitoes, S_v")
    title!(combined_plot[2, 1], "Susceptible mosquitoes, S_v")

    # Fourth subplot:
    for (i, I_v_opt) in enumerate(I_v_opts)
        plot!(combined_plot[2, 2], ts, I_v_opt,
            label="Scenario $(scenario_label[i])",
            linewidth=3, color=color_palette[i], thickness_scaling=1,
            xlim=(0, ts[end]),
            ylim=(0, 3500),
            xtickfontsize=12, ytickfontsize=12,
            xguidefontsize=14, yguidefontsize=14,
            legendfontsize=12, legend=:topright)
    end
    xlabel!(combined_plot[2, 2], "Time (days)")
    ylabel!(combined_plot[2, 2], "Infected mosquitoes, I_v")
    title!(combined_plot[2, 2], "Infected mosquitoes, I_v")

    # Fifth subplot:

```

```

    for (i, P_opt) in enumerate(P_opts)
        plot!(combined_plot[3, 1], ts, P_opt,
              label="Scenario $(scenario_label[i])",
              linewidth=3, color=color_palette[i], thickness_scaling=1,
              xlim=(0, ts[end]),
              ylim=(0, P_axis),
              xtickfontsize=12, ytickfontsize=12,
              xguidefontsize=14, yguidefontsize=14,
              legendfontsize=12, legend=(P_legend))
    end
    xlabel!(combined_plot[3, 1], "Time (days)")
    ylabel!(combined_plot[3, 1], "Partially immune, P")
    title!(combined_plot[3, 1], "Partially immune, P")

    if !isempty(u_opts)
        # u_opts = [u1_opt_1, u2_opt_1, u3_opt_1, u4_opt_1]
        for (i, u_opt) in enumerate(u_opts)
            plot!(combined_plot[3, 2], ts, u_opt, label="u$i", linewidth=3,
                  xlim=(0, ts[end]+1),
                  ylim=(0, 1.1),
                  xtickfontsize=12, ytickfontsize=12,
                  xguidefontsize=14, yguidefontsize=14,
                  legendfontsize=12, legend_columns=4,
                  legend=:topright)
        end
        xlabel!(combined_plot[3, 2], "Time (days)")
        title!(combined_plot[3, 2], "Control Profiles Scenario $(scenario_label[1])")
    end

    display(combined_plot)

    # Save the plot if `save=true`
    if save
        filename = "OptControl_Dengue_$((scenario_label[2])).png"
        savefig(combined_plot, filename)
        println("Plot saved as $filename")
    end
end;

```

To easily run the optimisation of different strategies

```

function run_optimization(u1_max, u2_max, u3_max, u4_max)

# Model definition with Ipopt optimizer
model = Model(Ipopt.Optimizer)
set_optimizer_attribute(model, "max_iter", 10000)
set_optimizer_attribute(model, "print_level", 5)
set_optimizer_attribute(model, "tol", 1e-6)

u1_init = 0
u2_init = 0
u3_init = 0
u4_init = 0

T = Int(tf/dt)

@variable(model, S_h[1:(T+1)] >= 0)      # Susceptible humans
@variable(model, I_h[1:(T+1)] >= 0)      # Infected symptomatic
@variable(model, I_hA[1:(T+1)] >= 0)    # Carriers asymptomatic
@variable(model, P[1:(T+1)] >= 0)       # Partially immune
@variable(model, R_h[1:(T+1)] >= 0)     # Recovered humans
@variable(model, S_v[1:(T+1)] >= 0)     # Susceptible mosquitoes
@variable(model, I_v[1:(T+1)] >= 0)     # Infected mosquitoes

@variable(model, 0 <= u1[1:(T+1)] <= u1_max) # Treated bednet
@variable(model, 0 <= u2[1:(T+1)] <= u2_max) # Vaccination
@variable(model, 0 <= u3[1:(T+1)] <= u3_max) # Treatment (prophylact
@variable(model, 0 <= u4[1:(T+1)] <= u4_max) # Insecticides

# Initial conditions
@constraints(model, begin
    S_h[1] == S_h0
    I_h[1] == I_h0
    I_hA[1] == I_hA0
    P[1] == P0
    R_h[1] == R_h0
    S_v[1] == S_v0
    I_v[1] == I_v0
    u1[1] == u1_init
    u2[1] == u2_init
    u3[1] == u3_init
    u4[1] == u4_init
    u1[T+1] == u1[T]

```

```

        u2[T+1] == u2[T]
        u3[T+1] == u3[T]
        u4[T+1] == u4[T]
    end);

    # Population sizes and infection rates
    @expressions(model, begin
        N_h[t=1:T], (S_h[t] + I_h[t] + I_hA[t] + P[t] + R_h[t])
        N_v[t=1:T], (S_v[t] + I_v[t])
        lambda_h[t=1:T], ((1 - u1[t]) * b * beta_1 / N_h[t]) * I_v[t]
        lambda_h1[t=1:T], ((1 - u1[t]) * b * beta_2 / N_h[t]) * I_v[t]
        lambda_v[t=1:T], (b * beta_3 / N_h[t]) * (I_h[t] + I_hA[t])
    end);

    # ODEs
    @constraints(model, begin
        [t=1:T], S_h[t+1] == S_h[t] + (mu_h * N_h[t] - lambda_h[t] * S_h[t] - S_h[t] *
        [t=1:T], I_h[t+1] == I_h[t] + (psi * lambda_h[t] * S_h[t] + omega * lambda_h[t] *
        [t=1:T], I_hA[t+1] == I_hA[t] + ((1 - psi) * lambda_h[t] * S_h[t] + (1 - omega) *
        [t=1:T], P[t+1] == P[t] + (u2[t] * S_h[t] + rho * u3[t] * I_h[t] + phi * gamma_h[t] *
        [t=1:T], R_h[t+1] == R_h[t] + ((1 - rho) * u3[t] * I_h[t] + (1 - phi) * gamma_h[t] *
        [t=1:T], S_v[t+1] == S_v[t] + (mu_v * N_v[t] * (1 - u4[t]) - lambda_v[t] * S_v[t] *
        [t=1:T], I_v[t+1] == I_v[t] + (lambda_v[t] * S_v[t] * (1 - u1[t]) - mu_v * I_v[t])
    end);

    # Objective function
    @objective(model, Min, sum(
        C_1 * I_h[t] + C_2 * I_hA[t] + C_3 * (S_v[t] + I_v[t]) + (D_1 * u1[t]^2
        for t in 1:T
    ));

    if silent
        set_silent(model)
    end

    # Run optimization
    optimize!(model)
    println("Termination status: $(termination_status(model))")

    return value.(u1), value.(u2), value.(u3), value.(u4), value.(S_h), value.(S_v), value.(I_h), value.(I_hA), value.(I_v), value.(P), value.(R_h);
end;

```

## Running the model without interventions

### Parameters

```
beta_1 = 0.75; #Transmission prob from I_v to S_h
beta_2 = 0.375; #Transmission prob from I_h to S_v
beta_3 = 0.75; #Transmission prb from I_v to P
b = 0.5; #Avg biting rate per mosquito per person
rho = 0.01; #Proportion of treated individuals with partial immunity
psi = 0.4; #Proportion of incidence rate from S_h to I_h
gamma_h = 0.3288330; #Disease related death rate of humans
omega = 0.54; #Proportion of incidence rate from P to I_h
mu_h = 0.0045; #Natural mortality rate and recruitment rate of humans
mu_v = 0.0323; #Natural mortality rate and recruitment rate of vector
phi = 0.48; #Proportion of natural Recovery
r_0 = 0.005; #Enhance death rate

u_1 = 0.0; #Treated bednet
u_2 = 0.0; #Vaccination
u_3 = 0.0; #Treatment (prophylactics)
u_4 = 0.0; #Insecticides
```

### Time and initial conditions

```
t0 = 0.0;
tf = 300.0;
dt = 0.5;
ts = collect(0:dt:tf);

S_h0 = 10000;
I_h0 = 100;
I_hA0 = 500;
P0 = 100;
R_h0 = 1000;
S_v0 = 6000;
I_v0 = 600;

delta = 0.001;

alg = Tsit5();
```



```
u0 = [S_h0, I_h0, I_hA0, P0, R_h0, S_v0, I_v0];
params = [beta_1, beta_2, beta_3, rho, psi, gamma_h, omega, mu_h, mu_v, phi, r_0];
```

```
prob_base = ODEProblem(dengue_ode!, u0, (t0, tf), params)
sol_base = solve(prob_base, alg, saveat=ts);
```

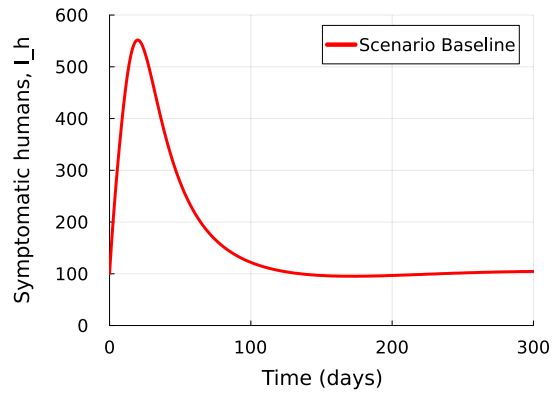
Extract data:

```
t_base = sol_base.t;
S_h_baseline = sol_base[1, :];
I_h_baseline = sol_base[2, :];
I_hA_baseline = sol_base[3, :];
P_baseline = sol_base[4, :];
R_h_baseline = sol_base[5, :];
S_v_baseline = sol_base[6, :];
I_v_baseline = sol_base[7, :];
```

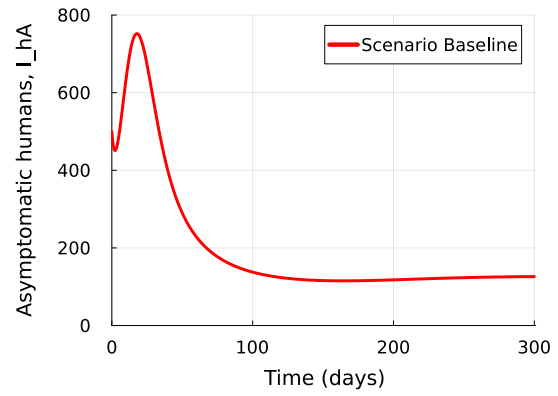
Plot baseline

```
Scenario_test = create_combined_plot(t_base,
[],
[S_h_baseline],
[I_h_baseline],
[I_hA_baseline],
[P_baseline],
[R_h_baseline],
[S_v_baseline],
[I_v_baseline],
["Baseline"], save=false)
```

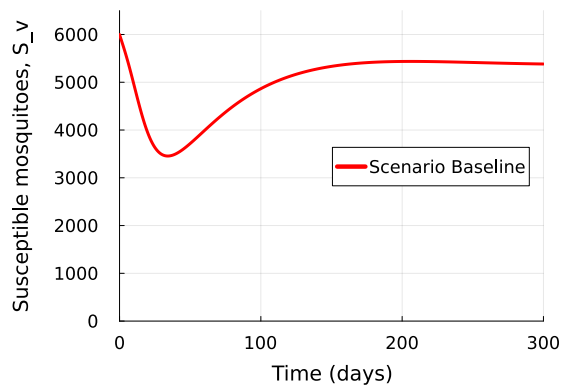
Symptomatic humans,  $I_h$



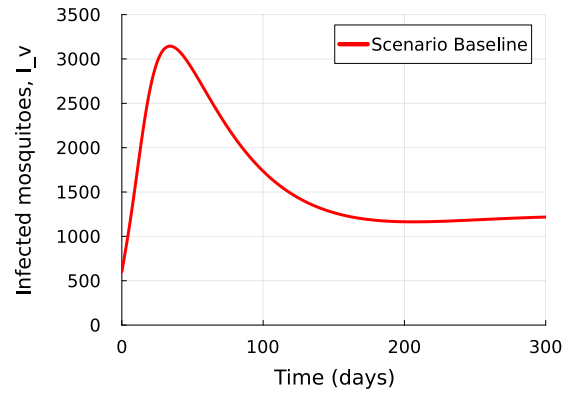
Asymptomatic humans,  $I_{hA}$



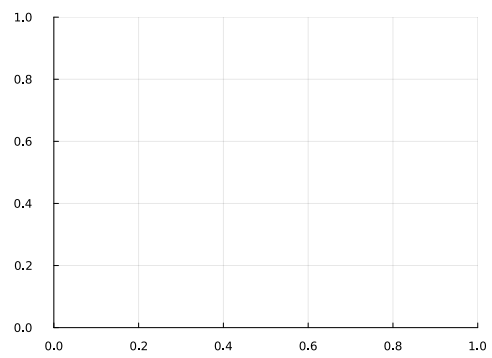
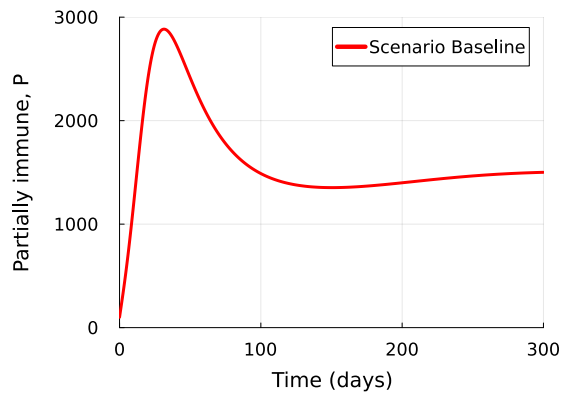
Susceptible mosquitoes,  $S_v$



Infected mosquitoes,  $I_v$



Partially immune,  $P$



## Strategies

```
# Weights for the objective function
C_1 = 5;
C_2 = 5;
C_3 = 5;
D_1= 16.62;
D_2 = 2.5;
D_3 = 5;
D_4 = 16.62;

silent = true;
```

### Strategy A

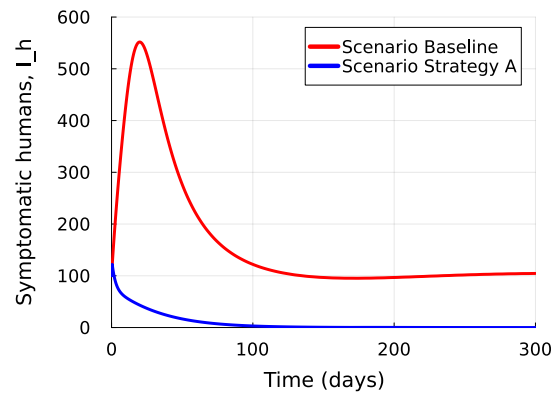
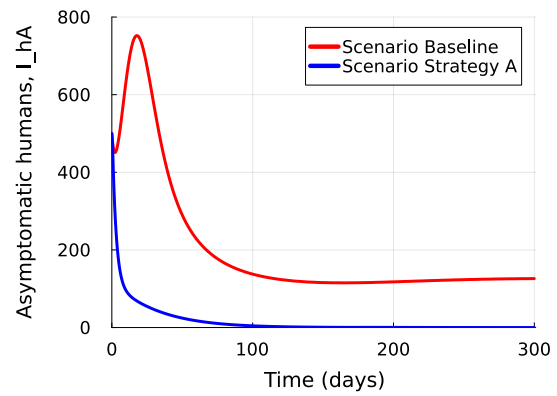
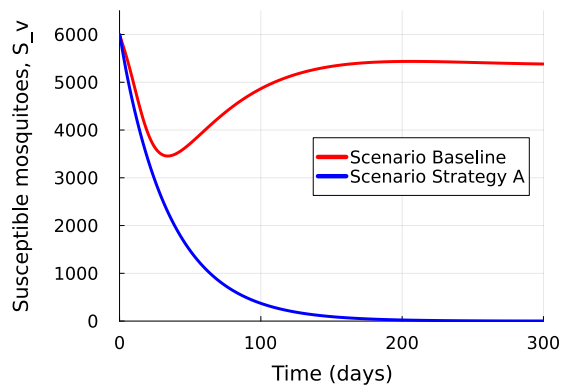
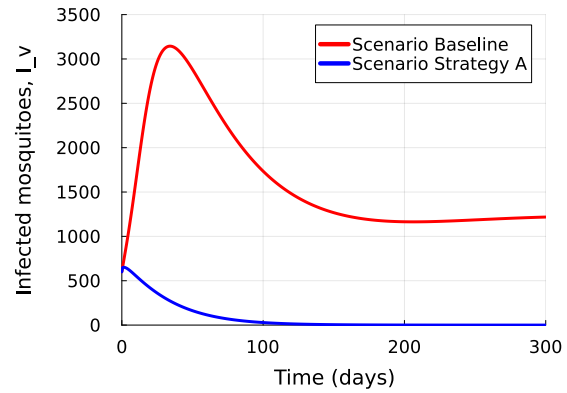
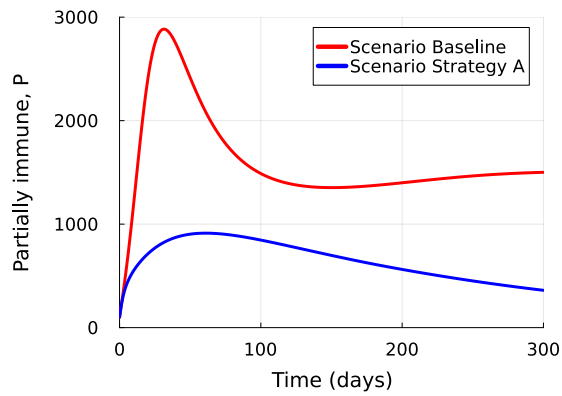
This strategy combines the use of treated bednets and insecticides

```
outputA_values = open("OptDengue_StratA_log.txt", "w") do file
  redirect_stdout(file) do
    return run_optimization(0.75, 0, 0, 0.75)
  end
end

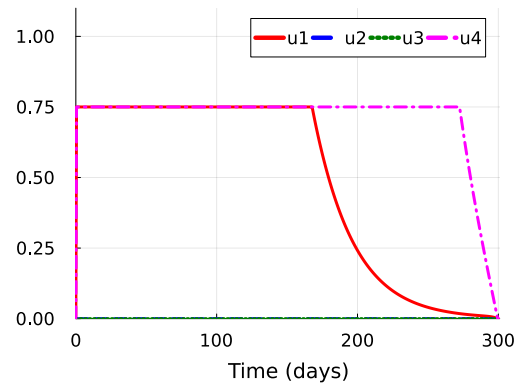
# Now extract values from output_values
u1_opt_A, u2_opt_A, u3_opt_A, u4_opt_A, S_h_opt_A, I_h_opt_A, I_hA_opt_A, P_opt_A,
```

Plot optimal results obtained from Strategy A vs Baseline

```
Scenario_A = create_combined_plot(t_base,
    [u1_opt_A, u2_opt_A, u3_opt_A, u4_opt_A],
    [S_h_baseline, S_h_opt_A],
    [I_h_baseline, I_h_opt_A],
    [I_hA_baseline, I_hA_opt_A],
    [P_baseline, P_opt_A],
    [R_h_baseline, R_h_opt_A],
    [S_v_baseline, S_v_opt_A],
    [I_v_baseline, I_v_opt_A],
    ["Baseline", "Strategy A"], save=false)
```

Symptomatic humans,  $I_h$ Asymptomatic humans,  $I_{hA}$ Susceptible mosquitoes,  $S_v$ Infected mosquitoes,  $I_v$ Partially immune,  $P$ 

Control Profiles Scenario Strategy A



## Strategy B

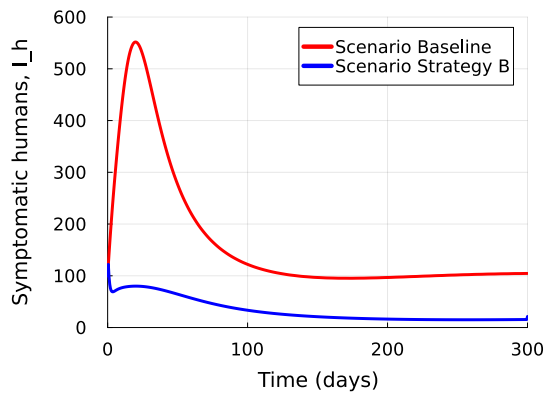
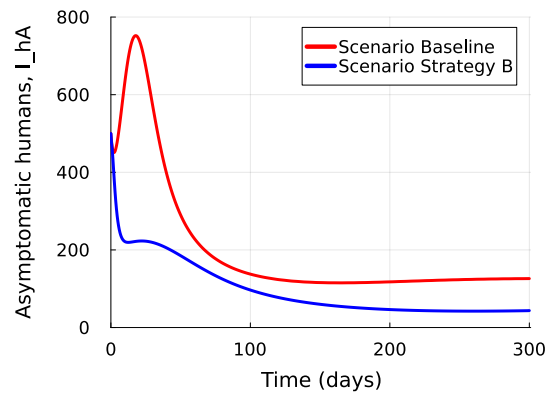
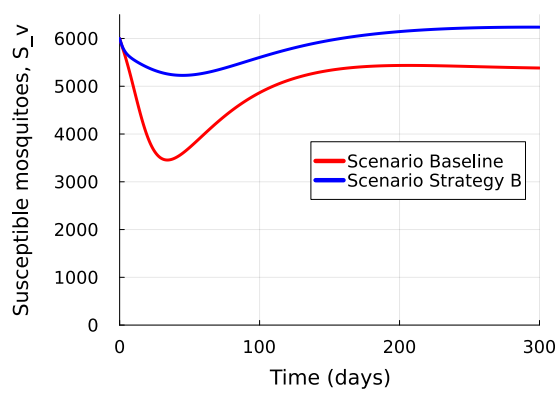
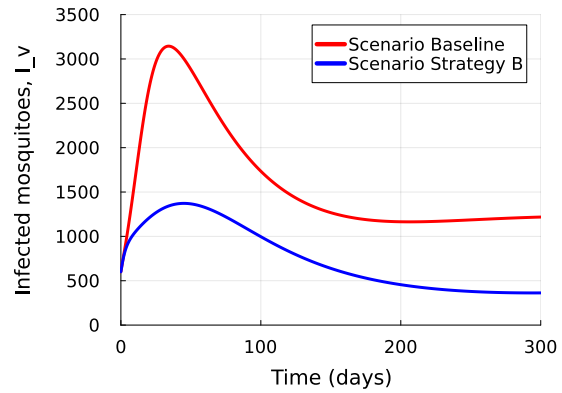
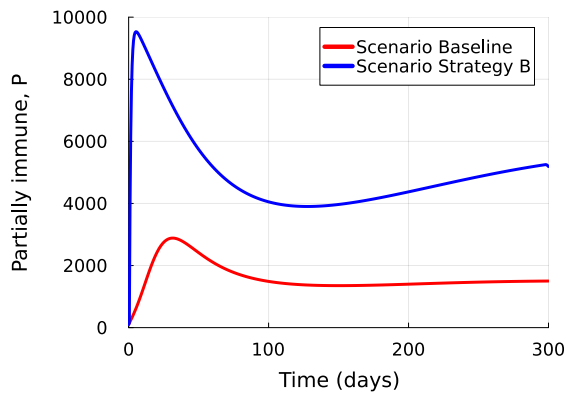
This strategy combines the use of treatment (prophylactics) and vaccination

```
outputB_values = open("OptDengue_StratB_log.txt", "w") do file
  redirect_stdout(file) do
    return run_optimization(0, 0.75, 0.75, 0)
  end
end

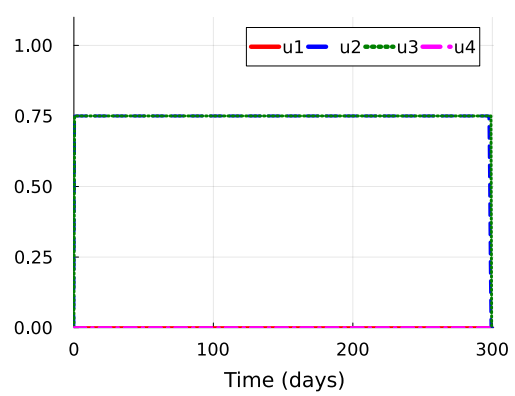
# Now extract values from output_values
u1_opt_B, u2_opt_B, u3_opt_B, u4_opt_B, S_h_opt_B, I_h_opt_B, I_hA_opt_B, P_opt_B
```

Plot optimal results obtained from Strategy B vs Baseline

```
Scenario_B = create_combined_plot(t_base,
  [u1_opt_B, u2_opt_B, u3_opt_B, u4_opt_B],
  [S_h_baseline, S_h_opt_B],
  [I_h_baseline, I_h_opt_B],
  [I_hA_baseline, I_hA_opt_B],
  [P_baseline, P_opt_B],
  [R_h_baseline, R_h_opt_B],
  [S_v_baseline, S_v_opt_B],
  [I_v_baseline, I_v_opt_B],
  ["Baseline", "Strategy B"], P_axis=10000, save=false)
```

Symptomatic humans,  $I_h$ Asymptomatic humans,  $I_{hA}$ Susceptible mosquitoes,  $S_v$ Infected mosquitoes,  $I_v$ Partially immune,  $P$ 

Control Profiles Scenario Strategy B



## Strategy C

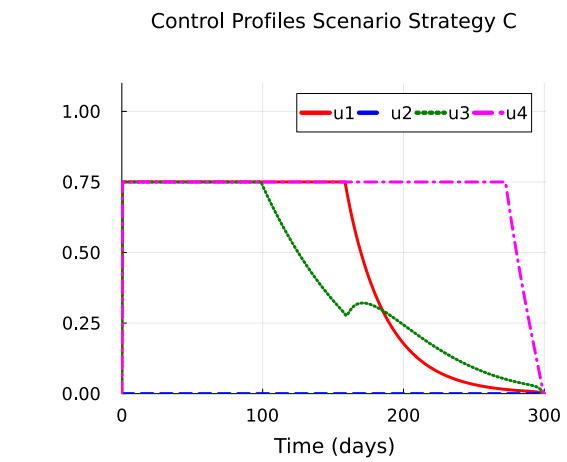
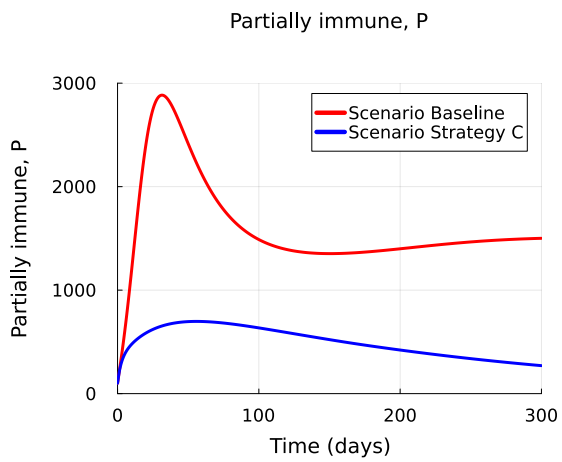
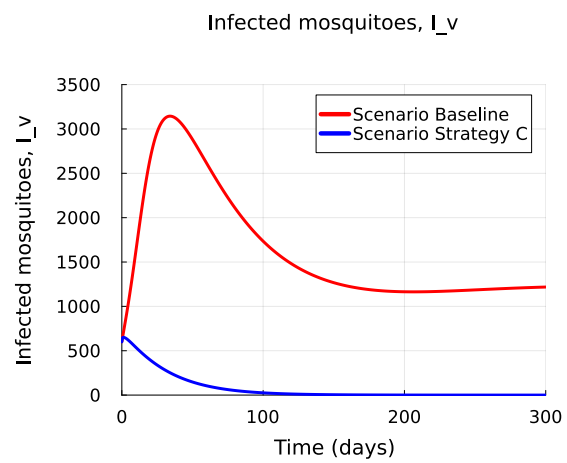
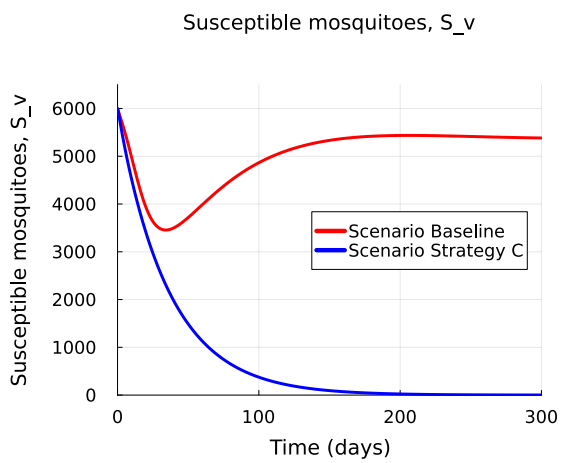
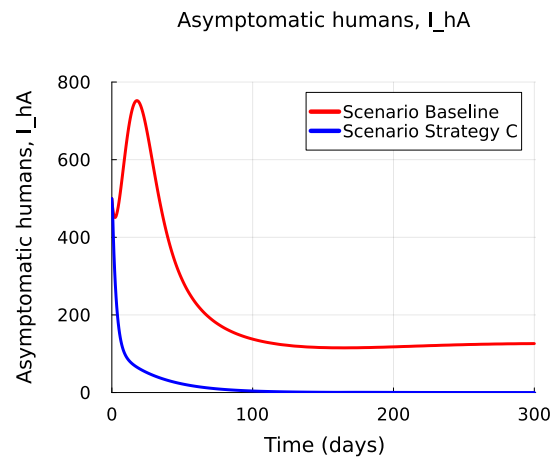
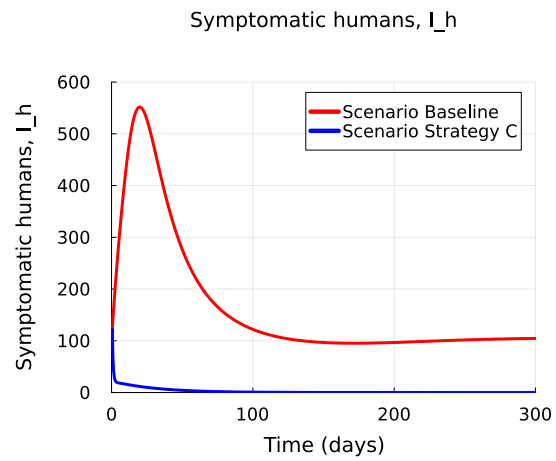
This strategy combines the use of treated bednets, treatment (prophylactics) and insecticides

```
outputC_values = open("OptDengue_StratC_log.txt", "w") do file
  redirect_stdout(file) do
    return run_optimization(0.75, 0, 0.75, 0.75)
  end
end

# Now extract values from output_values
u1_opt_C, u2_opt_C, u3_opt_C, u4_opt_C, S_h_opt_C, I_h_opt_C, I_hA_opt_C, P_opt_C
```

Plot optimal results obtained from Strategy C vs Baseline

```
Scenario_C = create_combined_plot(t_base,
  [u1_opt_C, u2_opt_C, u3_opt_C, u4_opt_C],
  [S_h_baseline, S_h_opt_C],
  [I_h_baseline, I_h_opt_C],
  [I_hA_baseline, I_hA_opt_C],
  [P_baseline, P_opt_C],
  [R_h_baseline, R_h_opt_C],
  [S_v_baseline, S_v_opt_C],
  [I_v_baseline, I_v_opt_C],
  ["Baseline", "Strategy C"], save=false)
```





## Strategy D

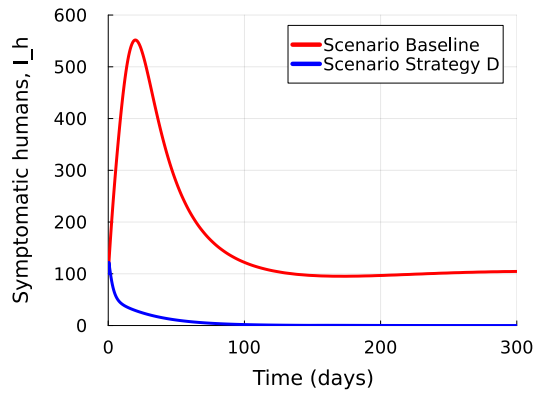
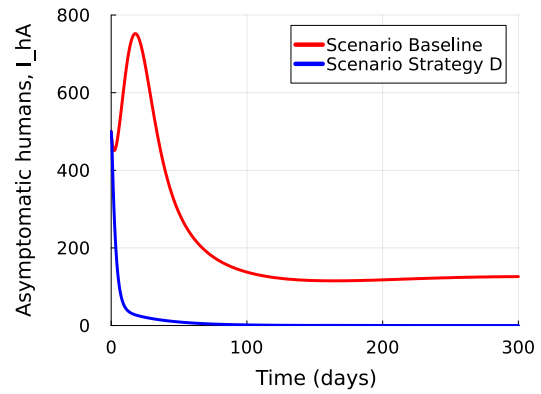
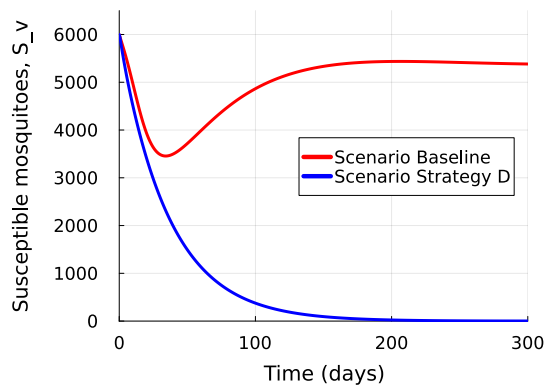
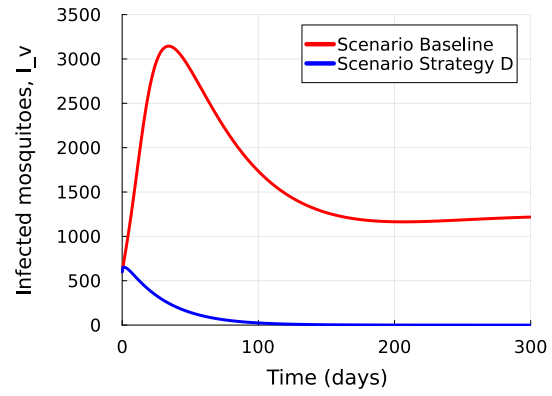
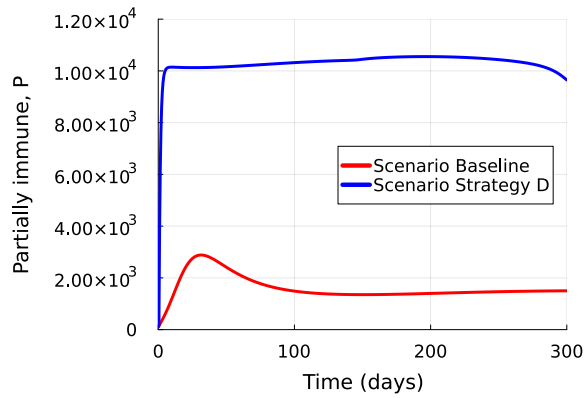
This strategy combines the use of treated bednets, insecticides and vaccination

```
outputD_values = open("OptDengue_StratD_log.txt", "w") do file
  redirect_stdout(file) do
    return run_optimization(0.75, 0.75, 0, 0.75)
  end
end

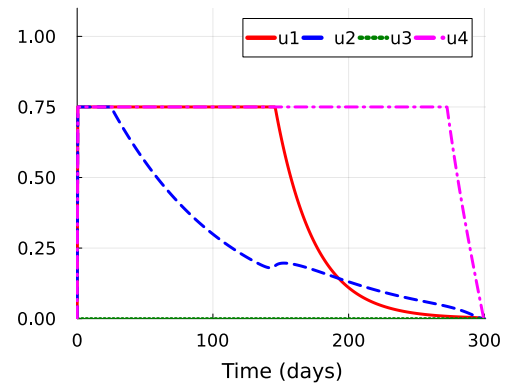
# Now extract values from output_values
u1_opt_D, u2_opt_D, u3_opt_D, u4_opt_D, S_h_opt_D, I_h_opt_D, I_hA_opt_D, P_opt_D,
```

Plot optimal results obtained from Strategy D vs Baseline

```
Scenario_D = create_combined_plot(t_base,
  [u1_opt_D, u2_opt_D, u3_opt_D, u4_opt_D],
  [S_h_baseline, S_h_opt_D],
  [I_h_baseline, I_h_opt_D],
  [I_hA_baseline, I_hA_opt_D],
  [P_baseline, P_opt_D],
  [R_h_baseline, R_h_opt_D],
  [S_v_baseline, S_v_opt_D],
  [I_v_baseline, I_v_opt_D],
  ["Baseline", "Strategy D"], P_axis=12000, P_legend=:right,
```

Symptomatic humans,  $I_h$ Asymptomatic humans,  $I_{hA}$ Susceptible mosquitoes,  $S_v$ Infected mosquitoes,  $I_v$ Partially immune,  $P$ 

Control Profiles Scenario Strategy D



## Strategy E

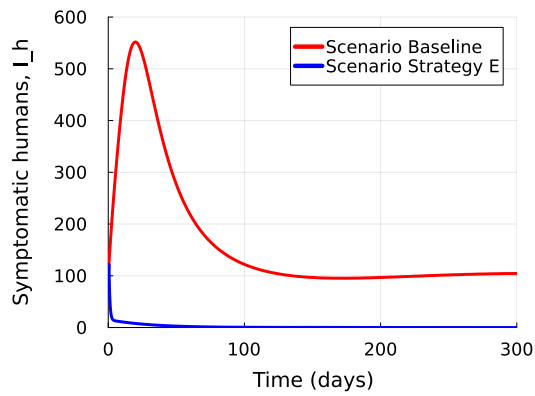
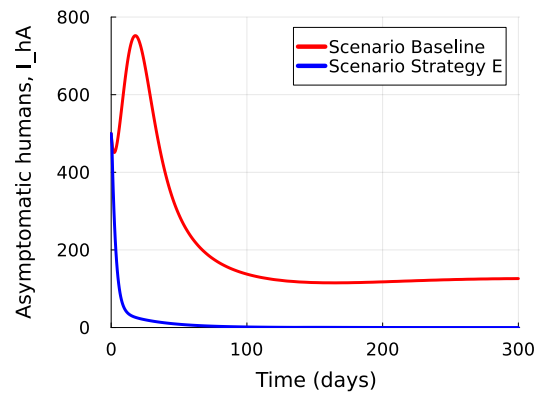
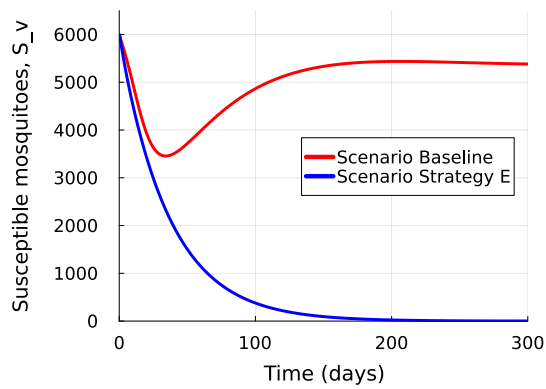
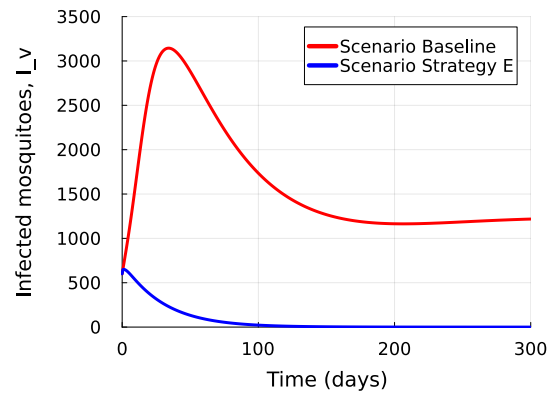
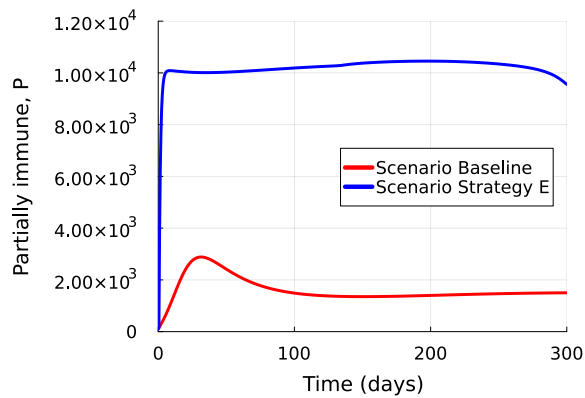
This strategy combines the use of the four control interventions treated bednets, insecticides, treatment and vaccination

```
outputE_values = open("OptDengue_StratE_log.txt", "w") do file
  redirect_stdout(file) do
    return run_optimization(0.75, 0.75, 0.75, 0.75)
  end
end

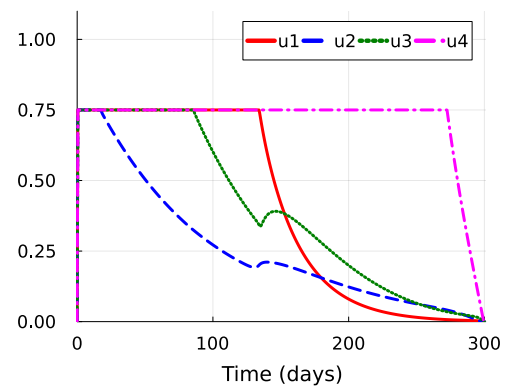
# Now extract values from output_values
u1_opt_E, u2_opt_E, u3_opt_E, u4_opt_E, S_h_opt_E, I_h_opt_E, I_hA_opt_E, P_opt_E
```

Plot optimal results obtained from Strategy E vs Baseline

```
Scenario_E = create_combined_plot(t_base,
  [u1_opt_E, u2_opt_E, u3_opt_E, u4_opt_E],
  [S_h_baseline, S_h_opt_E],
  [I_h_baseline, I_h_opt_E],
  [I_hA_baseline, I_hA_opt_E],
  [P_baseline, P_opt_E],
  [R_h_baseline, R_h_opt_E],
  [S_v_baseline, S_v_opt_E],
  [I_v_baseline, I_v_opt_E],
  ["Baseline", "Strategy E"], P_axis=12000, P_legend=:right,
```

Symptomatic humans,  $I_h$ Asymptomatic humans,  $I_{hA}$ Susceptible mosquitoes,  $S_v$ Infected mosquitoes,  $I_v$ Partially immune,  $P$ 

Control Profiles Scenario Strategy E



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